Hypothesis testing with linear models

DClin Research Methods 1

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Key points from Lecture 1:

- Generate research questions based on where the current literature is
- Incorporate the current theoretical models that explain your phenomenon of interest
- Some topics need more on the *“What?”* but many need more on the *“How?”* - build from wherever current knowledge is and design your study accordingly.
- Transparency and open science practices (e.g., registration) can prod you to clarify these points before you begin data collection.

Key points from Lecture 2:

- Statistical significance is an indicator of confidence that if the null hypothesis were true, we would be unlikely to find such a result.
- The importance of statistical significance is often ***overstated*** in that it does not:
  - Confirm that our alternative hypothesis is true
  - Mean that our result is clinically meaningful

The importance of statistical significance is often ***understated*** in that:

- There is no such thing as a “near significant” result - alpha thresholds (p values) are decided in advance
- Studies need to be designed with sufficient power (i.e., sample size) to detect the effect of interest
- The reliability of significant results depends on the analysis being applied correctly (e.g., test assumptions)

Key points from Lecture 2 (cont.):

- When designing studies, we need to understand the effect we are looking for and run power analysis
- When designing studies, we need to decide how best to assess the data to test our hypothesis/es
- When analysing the data, we need to look at it in detail, not just take a “cookier cutter” approach

Suggested reading:

The Eight Steps of Data Analysis: A Graphical Framework to Promote Sound Statistical Analysis

https://osf.io/preprints/psyarxiv/r8g7c

Overview

- Hypothesis testing
- Linear models
- Linear models in R

> Note: Some of the points discussed today will be best understood in combination with the next few lectures, but we have to introduce the concepts first.

Designing studies: think about the data!

When designing your study:

- Identify the most appropriate research question/hypothesis
- Figure out how the data you're collecting will help you test the hypothesis
- Figure out how best to model your hypothesis with the data you will have

> Do this at the design stage, when considering your method/measures. Don't wait until later!

Different types of research hypothesis

- Hypothesis testing is a method for making inferences about a population based on a sample
- In clinical psychology research, for example, we might be interested in the role of attentional bias in anxiety

Different types of research hypothesis

- However, we need to phrase this in terms of a hypothesis that we can test:
  - There is a difference in attentional bias between anxious and non-anxious individuals
  - There is a difference in attentional bias between anxious and non-anxious individuals, but only for threat-related stimuli
  - Level of anxiety predicts level of attentional bias to threat-related stimuli
  - Level of anxiety moderates the relationship between attentional bias and depression


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Hypothesis influences research design

- The nature of these hypotheses will determine the design of the study, the variables that are measured, and the statistical analysis that is used
- In many cases people come to analyse their data and find it difficult to know which statistical test to use
- This is not necessarily because of a lack of statistical knowledge

Different analyses for different designs?

Psych students are often taught to use different statistical tests for different types of designs. For example:

- t-test for comparing two groups
- ANOVA for comparing more than two groups
- Correlation for testing the relationship between two continuous variables
- Regression for testing the relationship between a continuous and a categorical variable
- ANCOVA for testing the relationship between a continuous and a categorical variable, controlling for a third continuous variable

All of the previous examples are based on the linear model and can be tested with regression

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Common statistical tests are linear models

Last updated: 28 June, 2019.Also check out the Python version!

See worked examples and more details at the accompanying notebook: https://lindeloev.github.io/tests-as-linear

	Common name	Built-in function in R	Equivalent linear model in R	Exact?	The linear model in words	Icon
Simple regression: lm(y ~ 1 + x)	**y is independent of x** P: One-sample t-test N: Wilcoxon signed-rank	t.test(y) wilcox.test(y)	lm(y ~ 1) lm(signed_rank(y) ~ 1)	✓ for N >14	One number (intercept, i.e., the mean) predicts **y**. - (Same, but it predicts the *signed rank* of **y**.)	
	P: Paired-sample t-test N: Wilcoxon matched pairs	t.test(y_1, y_2, paired=TRUE) wilcox.test(y_1, y_2, paired=TRUE)	lm(y_2 - y_1 ~ 1) lm(signed_rank(y_2 - y_1) ~ 1)	✓ for N >14	One intercept predicts the pairwise $\mathbf{y_2}$-$\mathbf{y_1}$ differences. - (Same, but it predicts the *signed rank* of $\mathbf{y_2}$-$\mathbf{y_1}$.)	
	y ~ continuous x P: Pearson correlation N: Spearman correlation	cor.test(x, y, method='Pearson') cor.test(x, y, method='Spearman')	lm(y ~ 1 + x) lm(rank(y) ~ 1 + rank(x))	✓ for N >10	One intercept plus **x** multiplied by a number (slope) predicts **y**. - (Same, but with *ranked* **x** and **y**)	
	y ~ discrete x P: Two-sample t-test P: Welch's t-test N: Mann-Whitney U	t.test(y_1, y_2, var.equal=TRUE) t.test(y_1, y_2, var.equal=FALSE) wilcox.test(y_1, y_2)	lm(y ~ 1 + G_2)^ gls(y ~ 1 + G_2, weights=…[¤])^ lm(signed_rank(y) ~ 1 + G_2)^	✓ ✓ for N >11	An intercept for **group 1** (plus a difference if **group 2**) predicts **y**. - (Same, but with one variance *per group* instead of one common.) - (Same, but it predicts the *signed rank* of **y**.)	
Multiple regression: lm(y ~ 1 + x_1 + x_2 +…)	P: One-way ANOVA N: Kruskal-Wallis	aov(y ~ group) kruskal.test(y ~ group)	lm(y ~ 1 + G_2 + G_3 +…+ G_N)^ lm(rank(y) ~ 1 + G_2 + G_3 +…+ G_N)^	✓ for N >11	An intercept for **group 1** (plus a difference if group ≠ 1) predicts **y**. - (Same, but it predicts the *rank* of **y**.)	
	P: One-way ANCOVA	aov(y ~ group + x)	lm(y ~ 1 + G_2 + G_3 +…+ G_N + x)^	✓	- (Same, but plus a slope on **x**.) *Note: this is discrete AND continuous. ANCOVAs are ANOVAs with a continuous x.*	
	P: Two-way ANOVA	aov(y ~ group * sex)	lm(y ~ 1 + G_2 + G_3 +…+ G_N + S_2 + S_3 +…+ S_K + G_2*S_2 + G_3*S_3 + … + G_N*S_K)	✓	Interaction term: changing **sex** changes the **y ~ group** parameters. *Note: G_{N-K} is an indicator (0 or 1) for each non-intercept levels of the **group** variable. Similarly for S_{2-K} for sex. The first line (with G_2) is main effect of group, the second (with S_2) for sex and the third is the **group × sex** interaction. For two levels (e.g. male/female), line 2 would just be "S_2" and line 3 would be S_2 multiplied with each G_2.*	*[Coming]*
	Counts ~ discrete x N: Chi-square test	chisq.test(groupXsex_table)	**Equivalent log-linear model** glm(y ~ 1 + G_2 + G_3 + … + G_N + S_2 + S_3 + … + S_K + G_2*S_2 + G_3*S_3 +…+ G_N*S_K, family=…)^	✓	Interaction term: (Same as Two-way ANOVA.) *Note: Run glm using the following arguments: `glm(model, family=poi(xson))` As linear-model, the Chi-square test is log(y_i) = log(N) + log(α_i) + log(β_i) + log($\alpha_i\beta_i$) where α_i and β_i are proportions. See more info in the accompanying notebook.*	*Same as Two-way ANOVA*
	N: Goodness of fit	chisq.test(y)	glm(y ~ 1 + G_2 + G_3 +…+ G_N, family=…)^	✓	(Same as One-way ANOVA and see Chi-Square note.)	*1W-ANOVA*

List of common parametric (P) non-parametric (N) tests and equivalent linear models. The notation y ~ 1 + x is R shorthand for $y = 1 \cdot b + a \cdot x$ which most of us learned in school. Models in similar colors are highly similar, but really, notice how similar they *all* are across colors! For non-parametric models, the linear models are reasonable approximations for non-small sample sizes (see "Exact" column and click links to see simulations). Other less accurate approximations exist, e.g., Wilcoxon for the sign test and Goodness-of-fit for the binomial test. The signed rank function is `signed_rank = function(x) sign(x) * rank(abs(x))`. The variables G_i and S_i are "dummy coded" indicator variables (either 0 or 1) exploiting the fact that when $\Delta x = 1$ between categories the difference equals the slope. Subscripts (e.g., G_2 or y_{-1}) indicate different columns in data. lm requires long-format data for all non-continuous models. All of this is exposed in greater detail and worked examples at https://lindeloev.github.io/tests-as-linear.

^ See the note to the two-way ANOVA for explanation of the notation.
¤ Same model, but with one variance per group: `gls(value ~ 1 + G2, weights = varIdent(form = ~1|group), method="ML")`.

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Jonas Kristoffer Lindeløv
https://lindeloev.net

Think about your questions before you think about the stats

The role of attentional bias in anxiety

Let’s assume we have several variables:

- Age, Gender
- Stimulus level: Threatening / Non-threatening
- Anxiety Score
- Attentional bias (based on reaction times)

How do we analyse the data?

In the coming weeks, we will see how taking a modelling approach allows us to answer the questions we want, rather than designing questions to fit a set of tests.

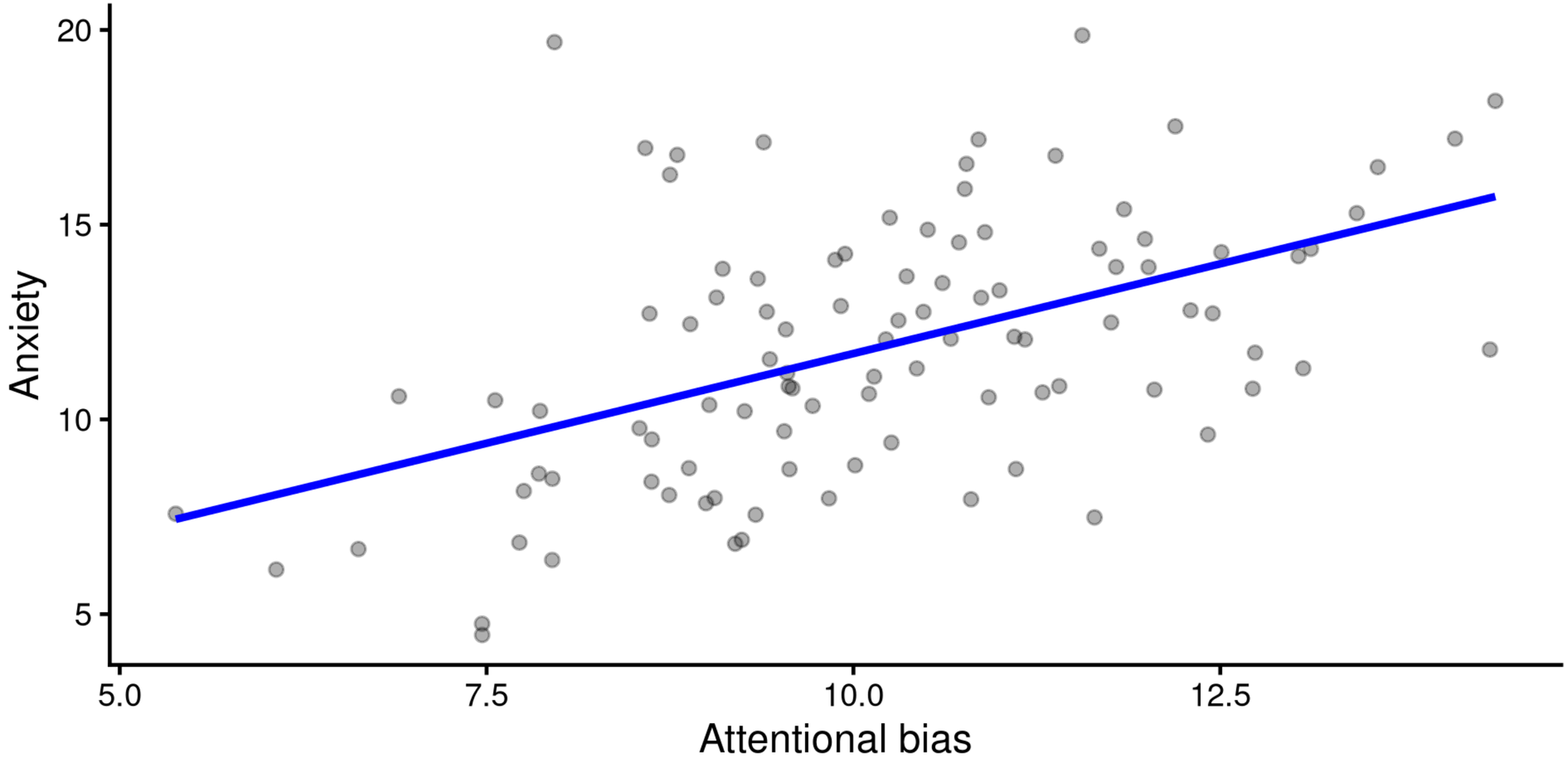
What does “modelling” the data actually mean?

- “Modelling” means figuring out what is the best way to represent our data.
- For research purposes, we could extend that to say “What is the best way to represent (and test) our hypothesis with the data?”

> For example, a t-test *models* the data of two groups as their means and their variance scores

Linear models using continuous predictor variables

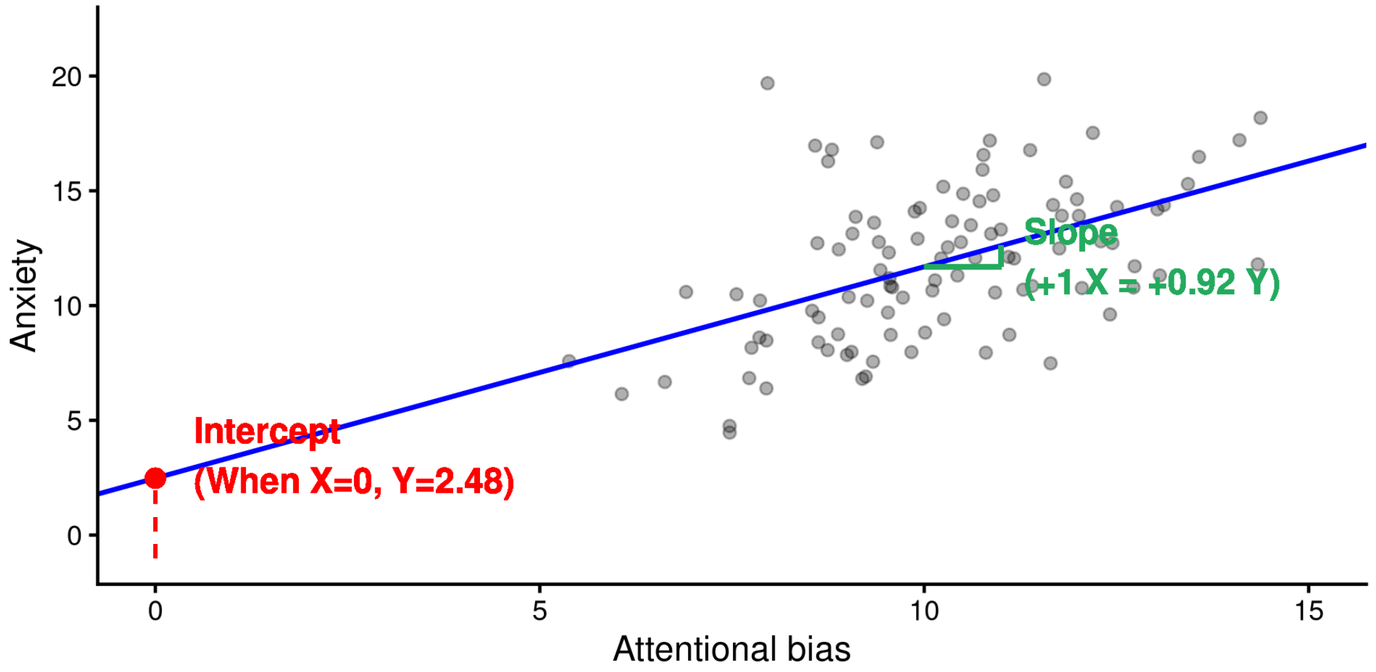
Linear regression



- The regression line is the line of best fit through the data (smallest amount of overall error between the line and the data points). Lowest possible sum of squared residuals.

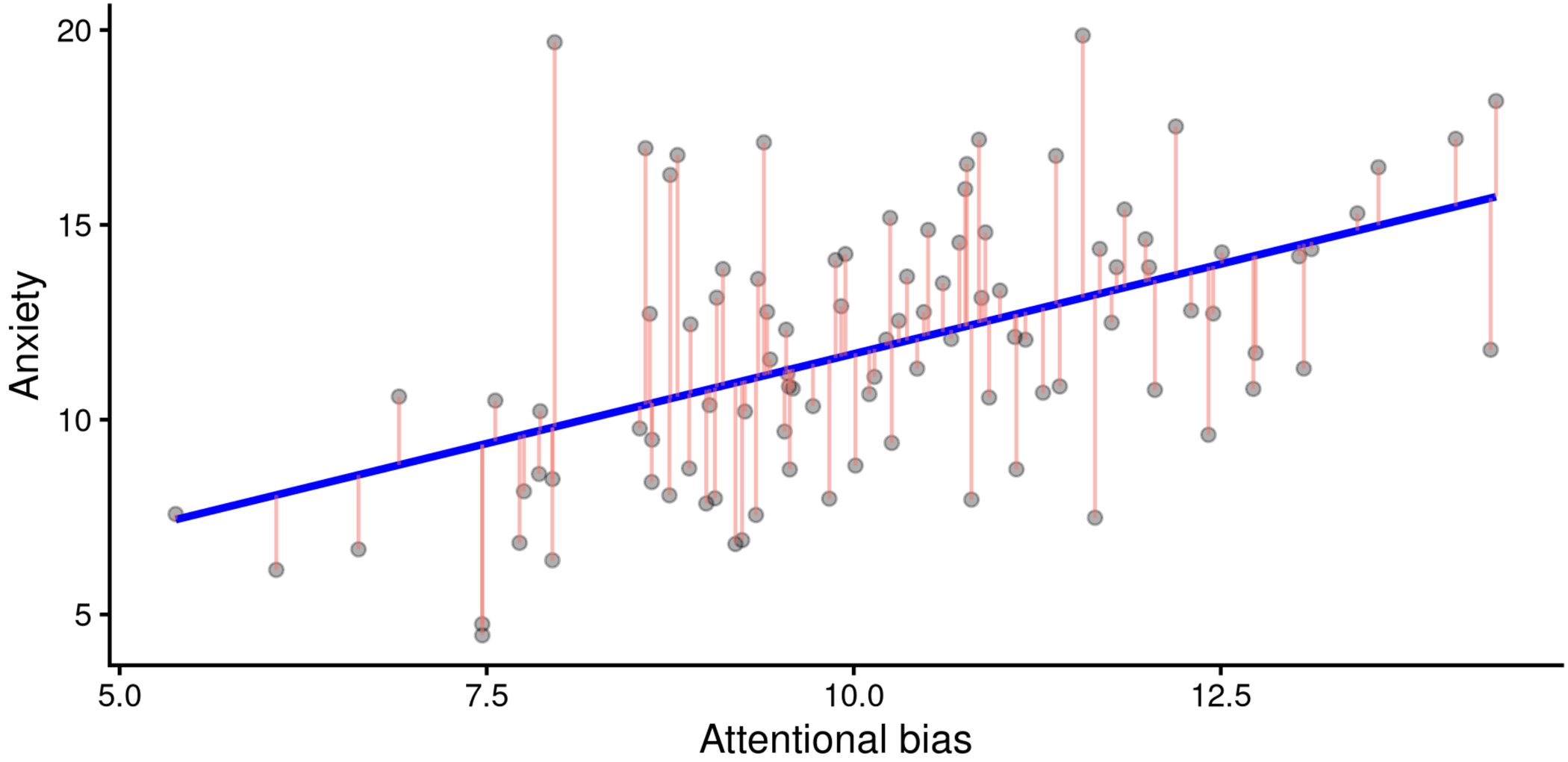
Visualizing the Intercept and Slope

- **Intercept (β_0):** The predicted value of y (Anxiety) when $x = 0$ (Attentional Bias).
- **Slope (β_1):** The predicted change in y for every 1-unit increase in x.



- The Intercept (Red): The predicted level of Anxiety when Attentional bias is exactly zero. Notice how the line must be traced backwards outside of our measured data range to find this value.
- The Slope (Green): The predicted change in Anxiety for every 1-unit increase in Attentional bias.

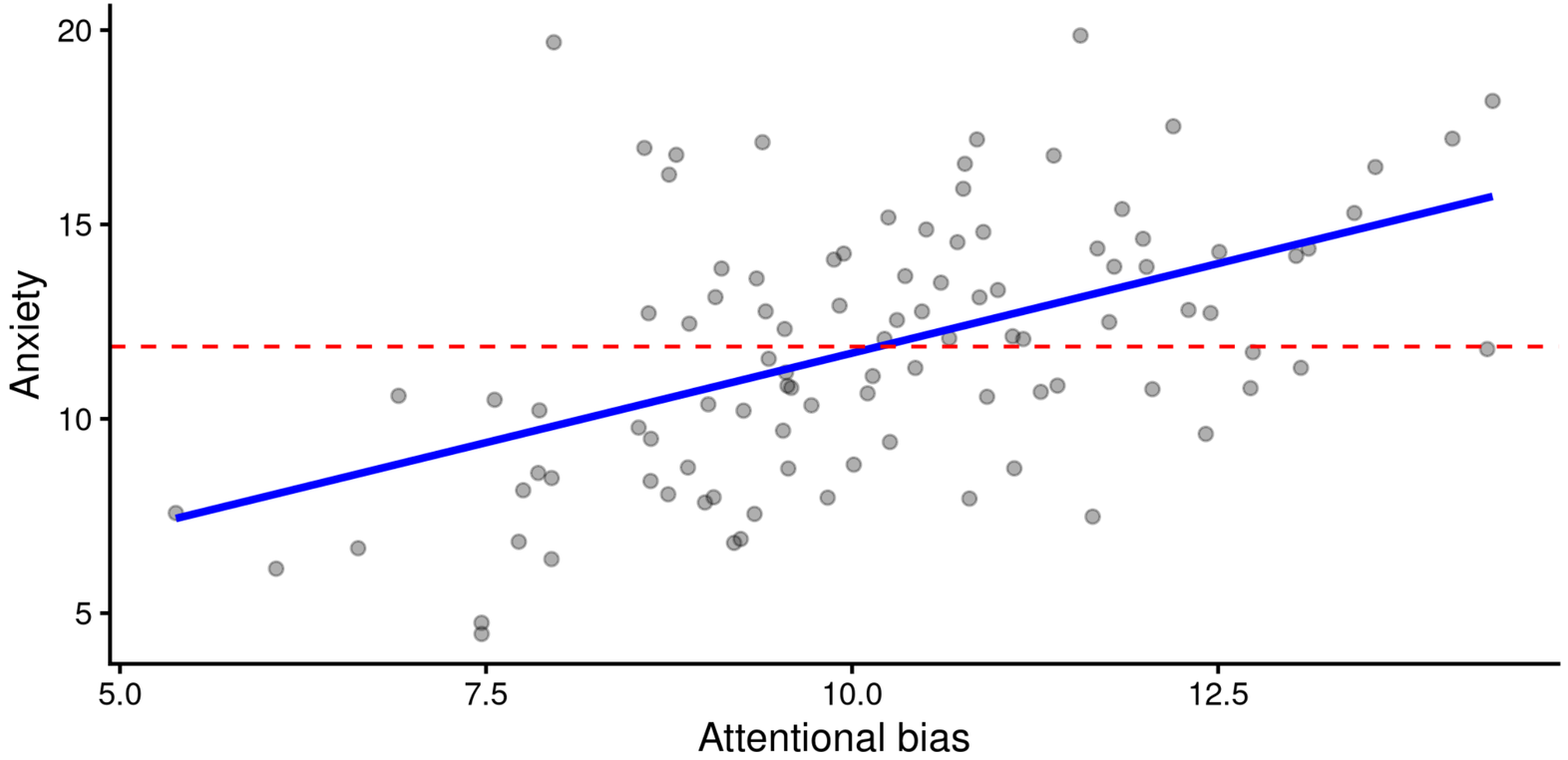
Residuals



- The distances between the data points and the regression line are the residuals

But what is the null hypothesis in regression?

The null hypothesis in regression



- The null hypothesis is that there is no relationship between the variables, meaning the slope of the regression line is exactly zero ($\beta_1 = 0$).
- A completely flat regression line is mathematically identical to a horizontal line drawn at the mean of the y variable. Therefore, the null hypothesis also posits that the predictor is no more useful than simply guessing the average y score for every participant.

Looking at regression output

```
1 model1 <- lm(y ~ x, data = df)                                    ①
2
3 summary(model1)                                                   ②
```

① The lm() function is used for linear regression. The model is specified using the formula y ~ x, where y is the outcome variable and x is the predictor variable.

② The summary() function is used to get a summary of the model. This includes the intercept and slope values, the standard errors, the t-values, and the p-values.

Looking at regression output

```r
model1 <- lm(y ~ x, data = df)

summary(model1)
```

```
Call:
lm(formula = y ~ x, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-5.7220 -2.0505 -0.2625  1.7419  9.8712

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.4787     1.6579   1.495    0.138
x             0.9213     0.1603   5.747 1.03e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

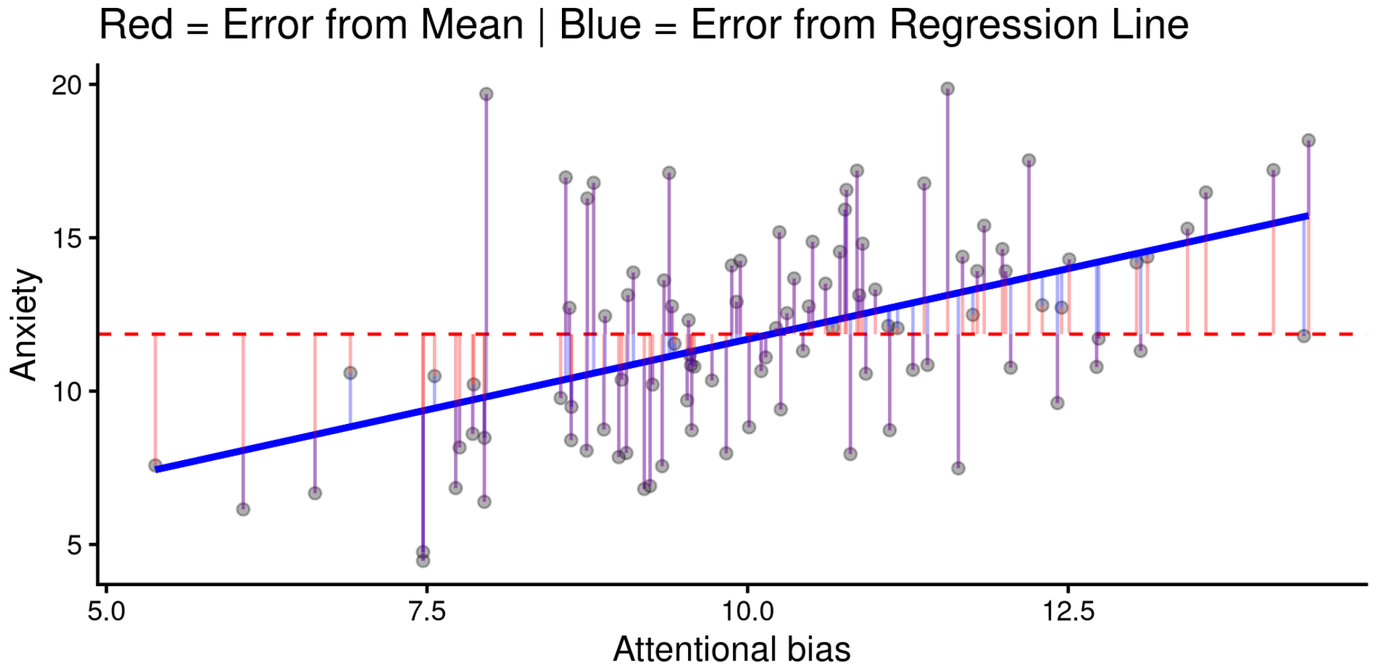
Residual standard error: 2.912 on 98 degrees of freedom
Multiple R-squared:  0.252, Adjusted R-squared:  0.2444
F-statistic: 33.02 on 1 and 98 DF,  p-value: 1.029e-07
```

Important elements:

- R^2**:** How much variance in the outcome variable is explained by the regression model?
- **model p-value:** Significance of the model (remember R output uses exponential notation)
- **predictor** β **value:** How much the outcome variable changes relative to a single unit increase in this variable
- β **p-value:** Significance of the individual predictor

Visualizing Model Fit (R^2)

- How do we know if our regression line is actually useful? We look at the variance it explains.
- **Total Error (Null Model):** The distance between the data points and the red dashed line (the mean of y).
- **Unexplained Error (Regression Model):** The distance between the data points and our blue regression line.
- R^2 **(R-squared):** The percentage of the original error that is eliminated by using the regression line instead of just guessing the mean.



What if we have a categorical predictor variable?

Example: t-test

- A t-test is a special case of a linear model where there is one predictor variable (group) and one outcome variable (DV)

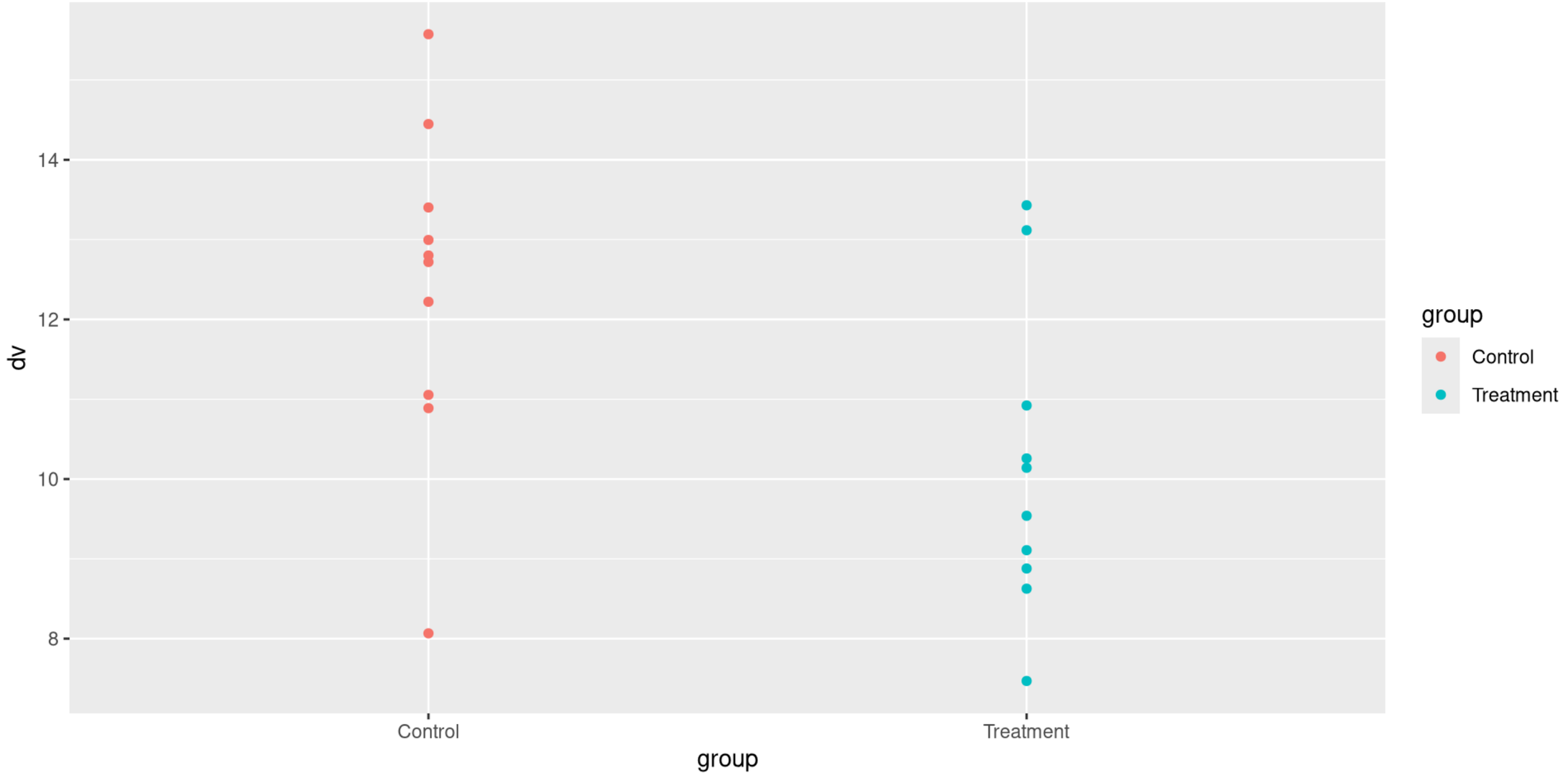
```
# run a t-test

t.test(dv ~ group, data = df)
```

```
    Welch Two Sample t-test

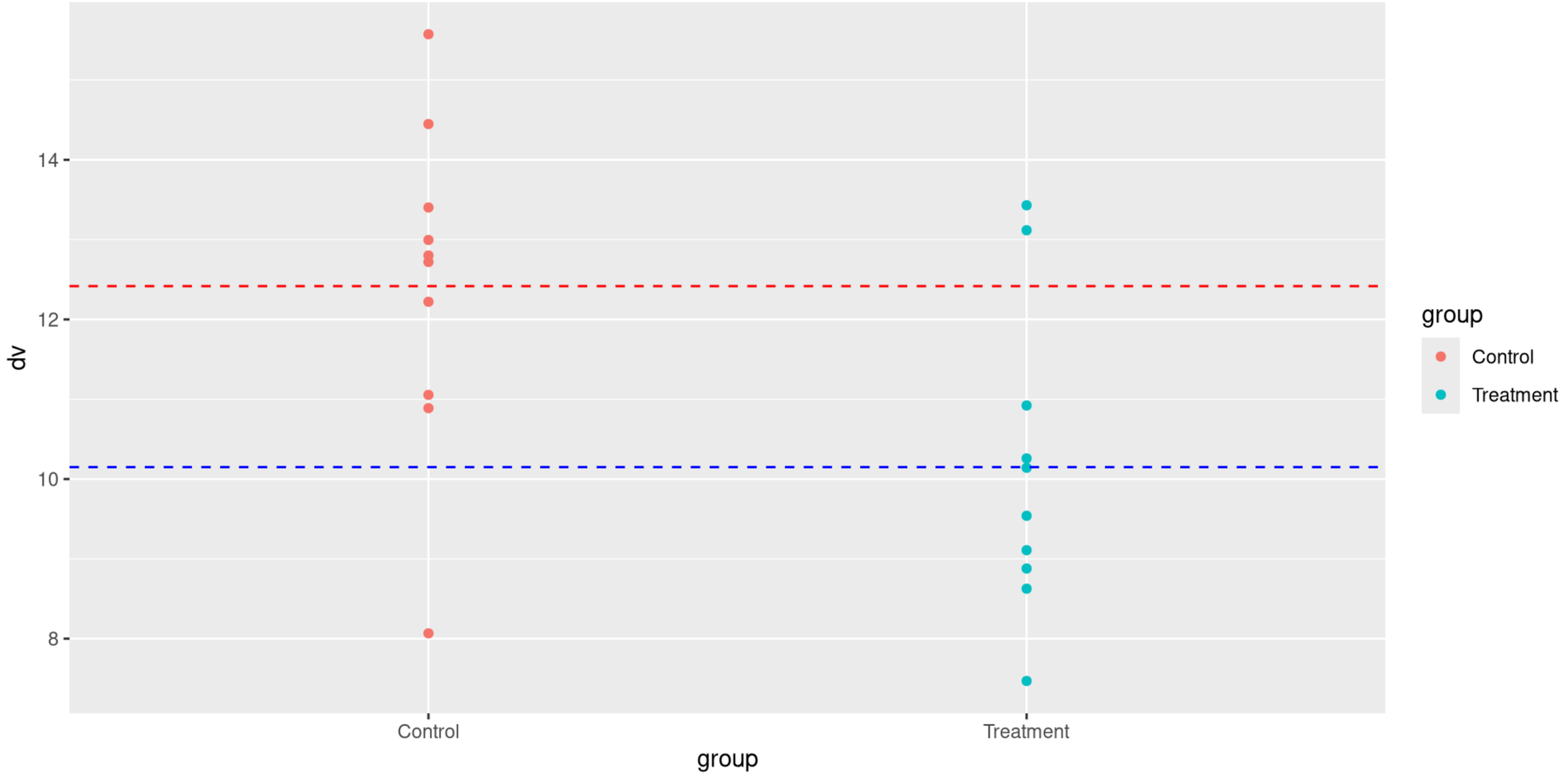
data:  dv by group
t = 2.5438, df = 17.872, p-value = 0.02044
alternative hypothesis: true difference in means between group Control and group Treatment is not equal to 0
95 percent confidence interval:
 0.3938876 4.1420977
sample estimates:
  mean in group Control mean in group Treatment
               12.41724                10.14925
```

Plotting the data #1



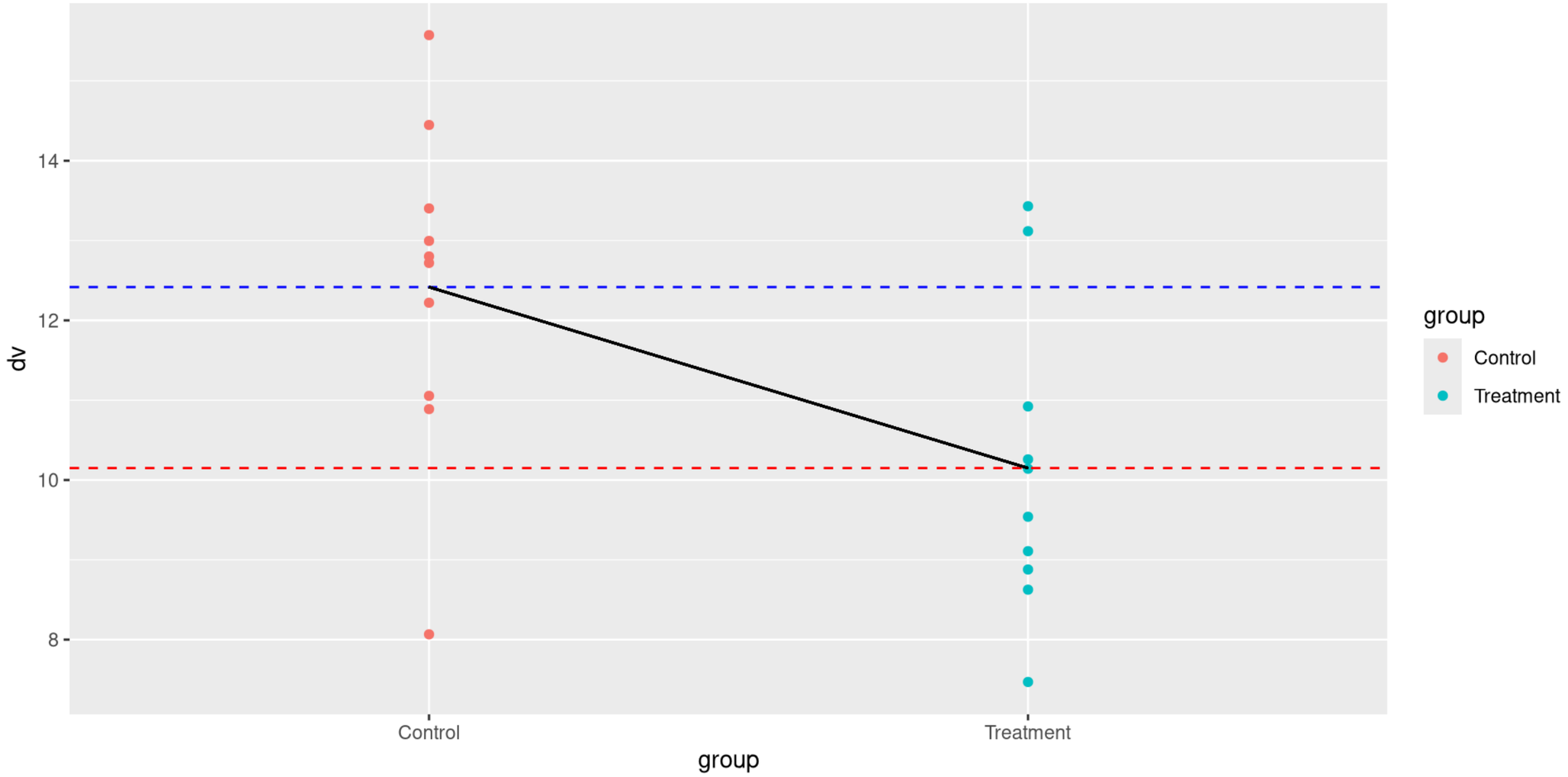
Here we can see the data for each group

Plotting the data #2



Here we can see the mean for each group, represented by the dashed lines

Plotting the data #3



In a regression model, the intercept is the mean of one of the groups, and the slope is the difference between the means of the two groups

Example: t-test as a linear model

```r
# run a linear model

lm(dv ~ group, data = df) %>% summary()
```

```
Call:
lm(formula = dv ~ group, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-4.3505 -1.2934  0.0505  0.8258  3.2809

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     12.4172     0.6304  19.696 1.25e-13 ***
groupTreatment  -2.2680     0.8916  -2.544   0.0204 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.994 on 18 degrees of freedom
Multiple R-squared:  0.2644,    Adjusted R-squared:  0.2236
F-statistic: 6.471 on 1 and 18 DF,  p-value: 0.02036
```

- In the above example, the t-test and linear model give the same results. The intercept for the linear model is the mean of the Control Group, and the slope is the difference between the means of group Control and Treatment.

Under the Hood: Dummy Coding

How does a linear model output a mean difference as a slope? R automatically applies **dummy coding** (0 and 1) to categorical variables.

Let's look at the regression equation: $Y = \beta_0 + \beta_1(X)$

If R codes our **Control Group as 0** and our **Treatment Group as 1**:

- **For a Control patient ($X = 0$):**
  - $Y = \text{Intercept} + \text{Slope} \times (0)$
  - $Y = \text{Intercept}$
  - *(Therefore, the Intercept MUST equal the mean of the Control group)*
- **For a Treatment patient ($X = 1$):**
  - $Y = \text{Intercept} + \text{Slope} \times (1)$
  - $Y = \text{Intercept} + \text{Slope}$
  - *(Therefore, the Slope MUST equal the exact difference between the Treatment and Control means)*

We can check the confidence intervals from the regression model

```r
# show the confidence confidence intervals of the coefficients

lm(dv ~ group, data = df) %>% confint()
```

```
                   2.5 %     97.5 %
(Intercept)    11.092729 13.7417583
groupTreatment -4.141139 -0.3948464
```

- We can see that the confidence interval of the regression coefficient is the same as the confidence interval of the difference between means in the t-test

What’s the value of using this approach?

Advantages of using models

- Using models allows us to test a wide range of hypotheses using the same approach
- This means that we can use the same approach to test hypotheses about:
  - the relationship between two continuous variables
  - the relationship between a categorical predictor and continuous outcome
  - Continuous and categorical predictors in the same model
- We can use this approach regardless of the number of predictor variables or levels in a categorical predictor

ANOVA as regression

One-way ANOVA

```r
# running an ANOVA

aov(dv ~ group, data = df) %>% summary()
```

```
            Df Sum Sq Mean Sq F value  Pr(>F)
group        2  48.97  24.486   6.435 0.00518 **
Residuals   27 102.73   3.805
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In this example, we can see that there is a significant effect of group on the outcome variable. However, we do not know which groups are significantly different from each other.

ANOVA as regression

```r
lm(dv ~ group, data = df) %>% summary()
```

```
Call:
lm(formula = dv ~ group, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-4.3505 -1.2824 -0.1021  0.9322  3.3567

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   10.1493     0.6168  16.454 1.34e-15 ***
groupGroup 2   2.2680     0.8723   2.600   0.0149 *
groupGroup 3   3.0016     0.8723   3.441   0.0019 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

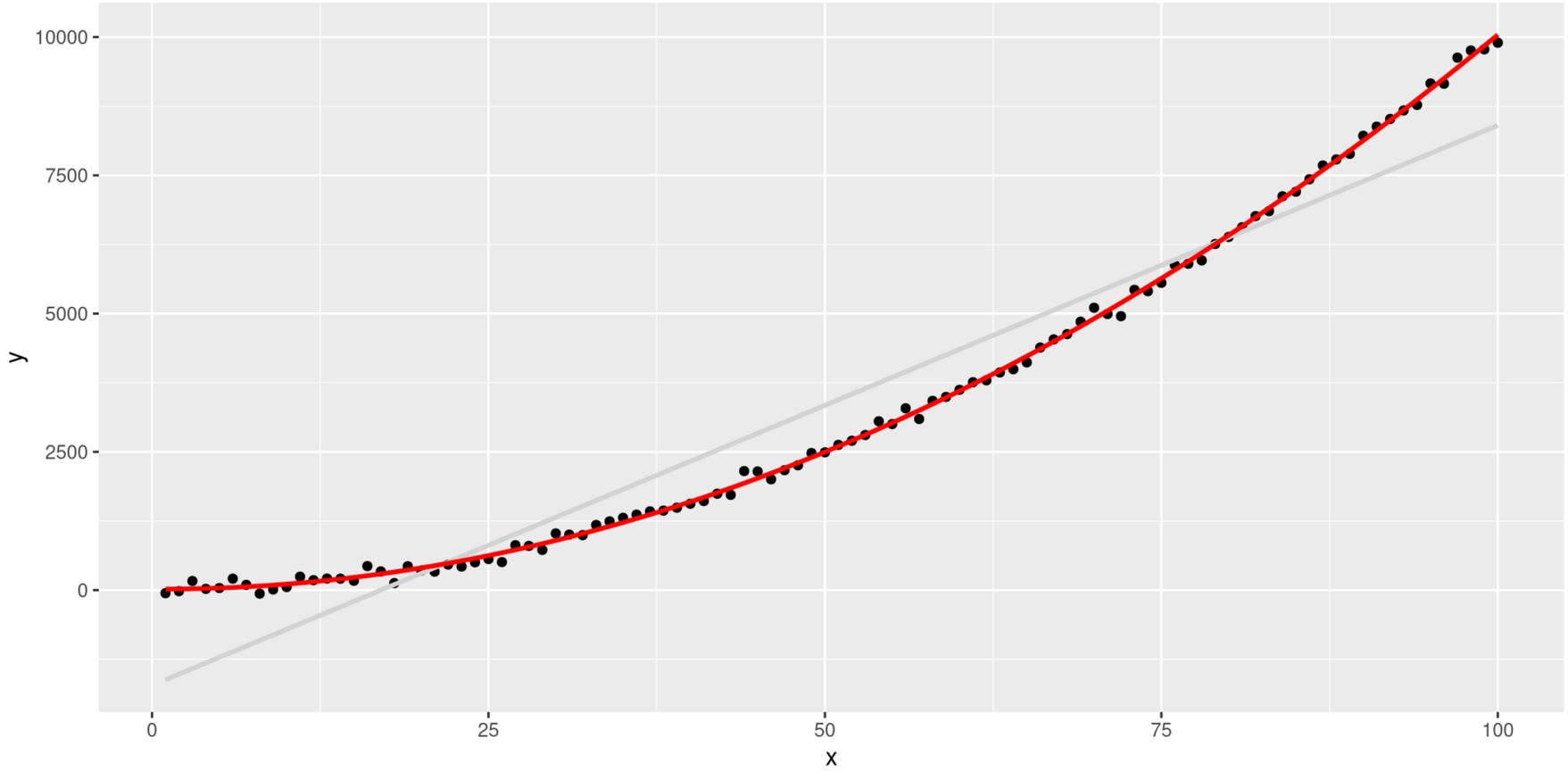
Residual standard error: 1.951 on 27 degrees of freedom
Multiple R-squared:  0.3228,    Adjusted R-squared:  0.2727
F-statistic: 6.435 on 2 and 27 DF,  p-value: 0.005183
```

In this example, we can see that there is a significant effect of group on the outcome variable. However, we can also see:

- that Group 2 and Group 3 are significantly different from Group 1.
- the exact difference between the means of each group

What about non-linear relationships?

- Not all relationships between variables are linear
- For example, the relationship between age and cognitive functioning is often non-linear
- In these cases, we can use an adapted linear approach (e.g., polynomial regression) to test the relationship



What about non-linear relationships? #2

- Once you become confident with the idea of modelling the relationships between variables, you can apply different types of model, based on what fits the data
- **Fit** is an important concept in modelling. *For example, to test a binary outcome (e.g., Yes/No), we would use logistic regresssion, because this type of model fits the data better.*
- There are also ways to model data to fit non-linear relationships for continuous data
- We are starting with the linear model, since all of the alternative approaches you know (e.g., t-test, ANOVA etc.) are based on a linear model.

How do we know if a linear model is appropriate?

- You need to check the assumptions of linear models when using them to test hypotheses
- These assumptions are:
  - Linearity
  - Independence
  - Homoscedasticity
  - Normality
- Many of these assumptions can be checked using the residuals of the model. If you are using R, the plot() function can be used to check these assumptions. We will learn another method next week

Summary

- Once your research question has been decided, figure out how you can model your hypothesis with the data you are collecting
- Many different types of hypothesis can be tested using regression models
- This can allow us to ask questions that are more complex because we can include multiple predictor variables in the same model (next week)
- We can get more information from a regression output than from a t-test or ANOVA (for example)
- However, we need to check the assumptions of linear models before reporting them